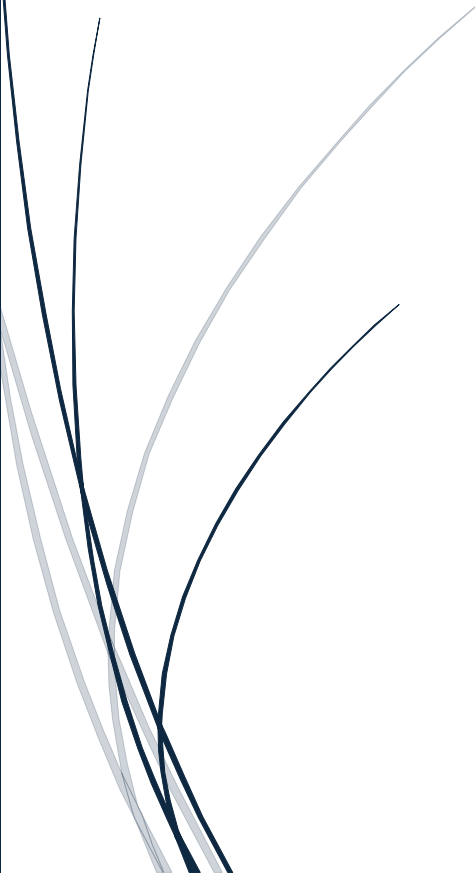




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E-Commerce Sales Analysis

Internship Project Report



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1. Introduction

The objective of this project was to analyze e-commerce sales data to identify top-performing product categories, underperforming regions, and seasonal sales patterns. The analysis was conducted using Python, Pandas, Matplotlib, and Power BI to support business decision-making through data-driven insights.

2. Dataset Overview

The project used the Brazilian E-Commerce (Olist) dataset containing information related to customers, orders, products, and order items.

Datasets Used

- olist_customers_dataset.csv
- olist_orders_dataset.csv
- olist_order_items_dataset.csv
- olist_products_dataset.csv
- product_category_name_translation.csv

Key Attributes

- Customer Information
 - Order Details
 - Product Categories
 - Product Prices
 - Order Status
 - Purchase Dates
 - Customer States
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3. Data Cleaning and Preparation

Several preprocessing steps were performed to ensure data quality.

Activities Performed

- Checked missing values using Pandas.
- Removed records with missing product category information.
- Converted date columns into datetime format.
- Merged multiple datasets using common keys such as order_id, customer_id, and product_id.
- Validated data consistency and removed duplicate records where necessary.

Missing Values Observed

- Product category information had a small number of missing records.
- Delivery-related columns contained missing values due to canceled or undelivered orders.
- Missing records were handled appropriately based on business context.

4. Exploratory Data Analysis (EDA)

Product Category Analysis

Revenue was calculated for each product category to identify the highest and lowest performing categories.

Top Revenue Categories

1. Health & Beauty
2. Watches & Gifts
3. Bed Bath & Table
4. Sports & Leisure
5. Computer Accessories

Bottom Revenue Categories

1. Insurance & Services
2. Kids Fashion
3. CDs & DVDs

4. Flowers
5. PC Gamer

Regional Performance Analysis

Customer state information was used to evaluate regional sales performance.

Top Performing States

1. São Paulo (SP)
2. Rio de Janeiro (RJ)
3. Minas Gerais (MG)
4. Rio Grande do Sul (RS)
5. Paraná (PR)

Underperforming States

1. Roraima (RR)
2. Amapá (AP)
3. Acre (AC)
4. Amazonas (AM)
5. Rondônia (RO)

Seasonal Trend Analysis

Monthly sales revenue was analyzed using order purchase dates.

Key observations include:

- Revenue increased steadily during 2017 and 2018.
 - November 2017 recorded the highest monthly revenue.
 - Strong sales performance was observed during holiday and promotional periods.
 - Sales remained consistently high during early 2018.
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5. Dashboard Development

An interactive Power BI dashboard was developed to visualize business performance.

Dashboard Components

- Total Revenue KPI
- Total Orders KPI
- Total Customers KPI
- Average Order Value KPI
- Delivered Orders Percentage KPI
- Monthly Revenue Trend
- Top 10 Product Categories
- Bottom 10 Product Categories
- Top 10 States by Revenue
- Bottom 10 States by Revenue
- Order Status Distribution
- Management Recommendations

The dashboard allows users to monitor sales performance and identify opportunities for business growth.

6. Key Business Insights

Insight 1

Health & Beauty products generated the highest revenue, indicating strong customer demand.

Insight 2

Revenue is concentrated among a small number of product categories, while several categories contribute very little to total sales.

Insight 3

São Paulo is the strongest market and contributes significantly more revenue than any other state.

Insight 4

November 2017 recorded the highest monthly revenue, highlighting seasonal purchasing behavior.

Insight 5

Northern states such as RR, AP, and AC show comparatively low revenue and may require additional business focus.

7. Recommendations

1. Increase marketing activities in underperforming states such as RR, AP, and AC.
2. Focus inventory and promotional campaigns on high-performing categories such as Health & Beauty and Watches & Gifts.
3. Prepare inventory in advance for peak demand periods, especially during November and early-year months.
4. Expand business operations in high-performing states including SP, RJ, and MG.
5. Review low-performing categories and optimize product offerings based on customer demand.

8. Conclusion

This project successfully analyzed e-commerce sales performance using Python and Power BI. The analysis identified key revenue-generating categories, regional sales trends, and seasonal demand patterns. The developed dashboard provides actionable insights that can support strategic business decisions, improve sales performance, and enhance customer engagement.

The findings demonstrate the value of data analytics in understanding customer behavior and driving business growth.